

Probing Warmth and Competence Representations in LLM Hiring Decisions

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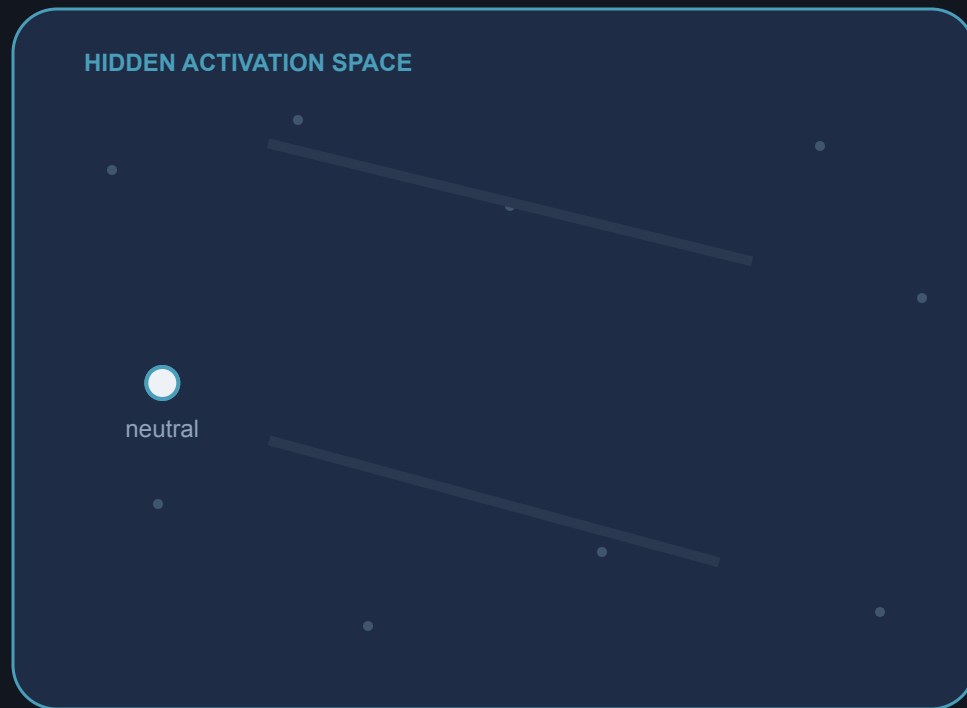
AI is now screening job applications, and it discriminates



All existing audits stop here: they see the output, not the mechanism

- 50% of US organizations use AI in recruiting (SHRM, 2025)
- Behavioral audits show name-driven disparities across race and gender
- But they treat the model as a black box: that it discriminates, not how

A story becomes a hidden activation



A story becomes a hidden activation

STORY

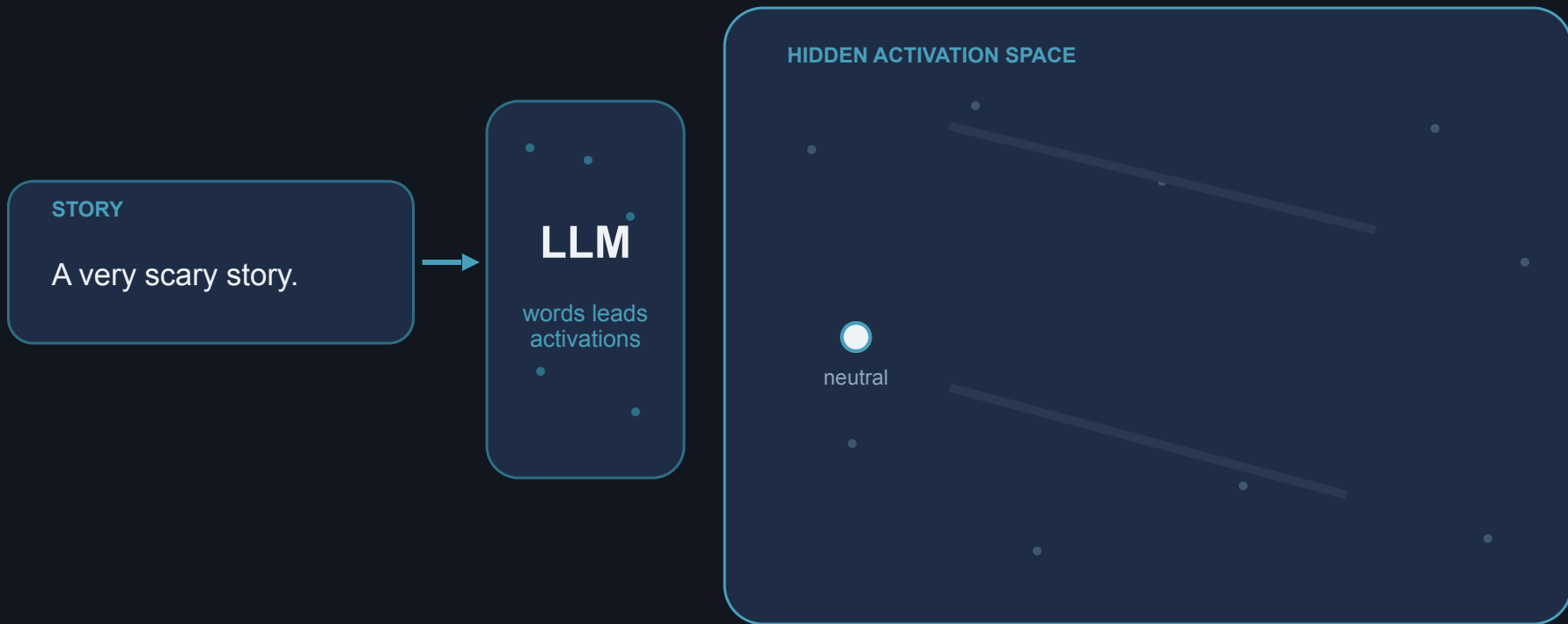
A very scary story.

HIDDEN ACTIVATION SPACE

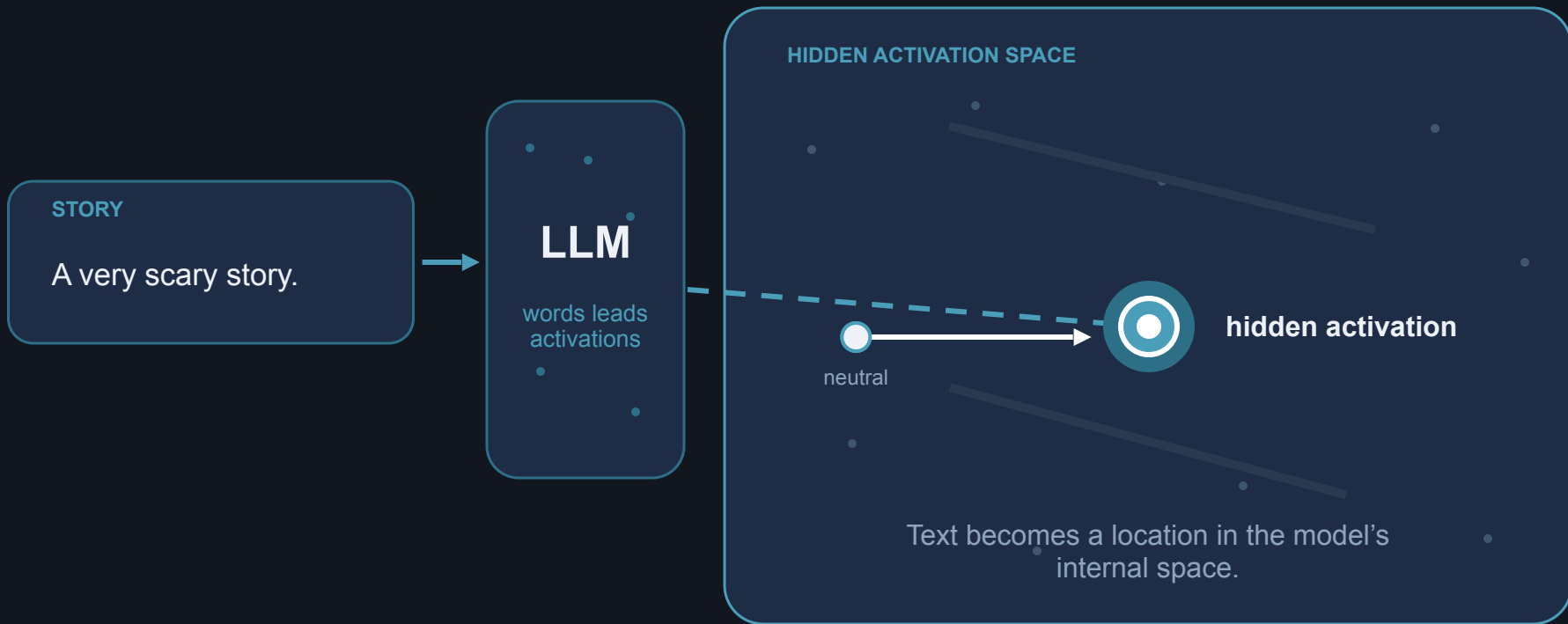
neutral



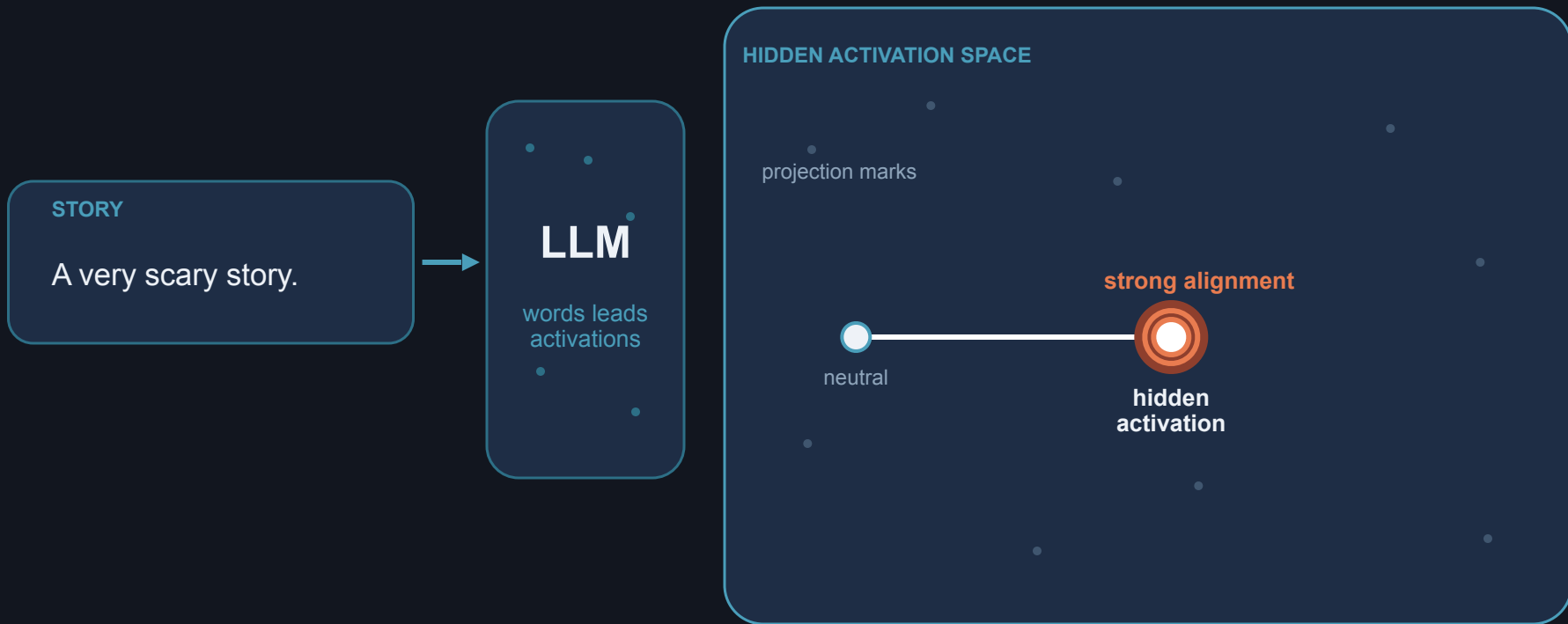
A story becomes a hidden activation



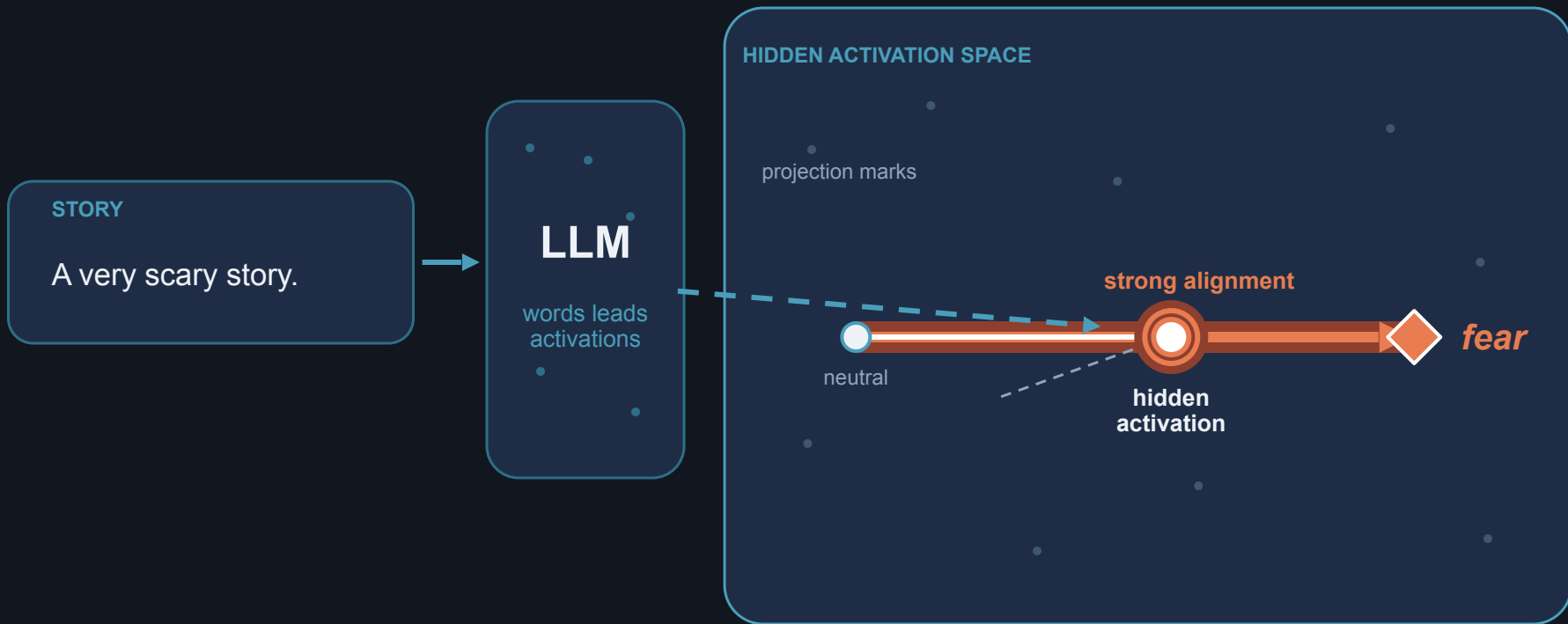
A story becomes a hidden activation



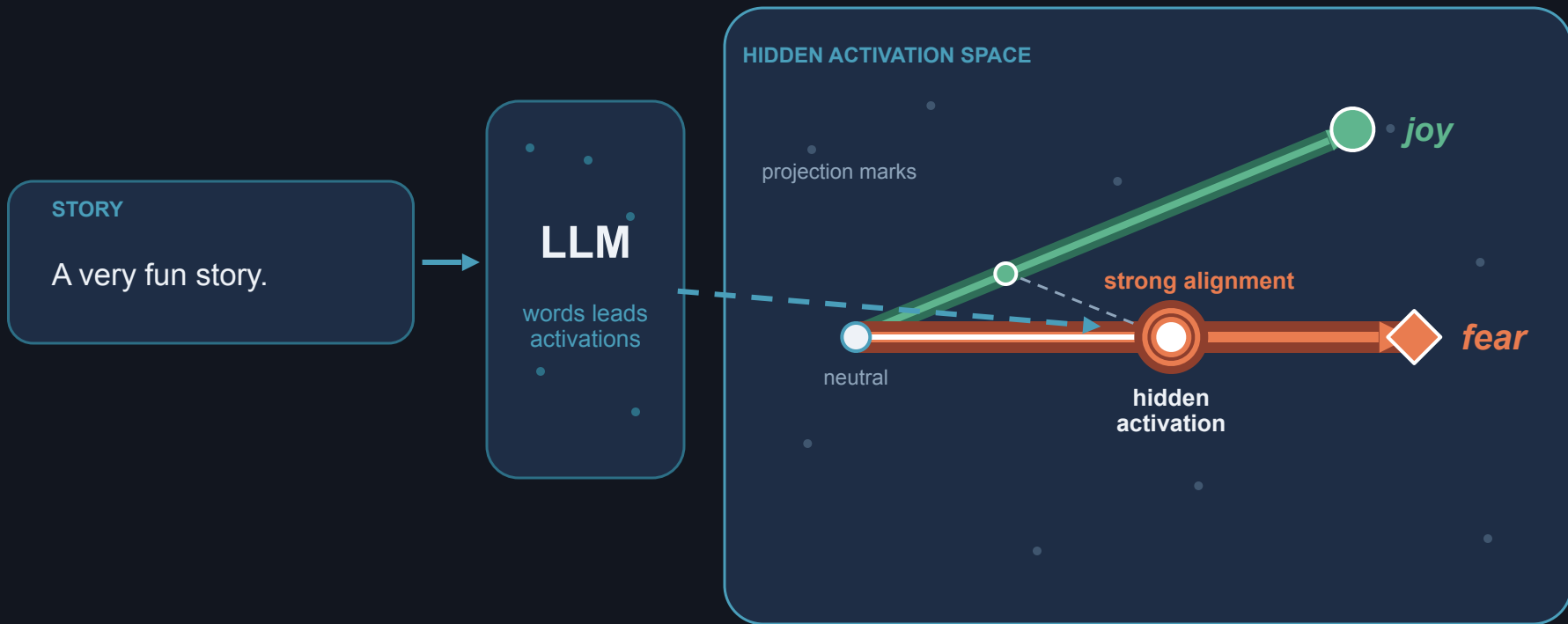
Emotion concepts become directions



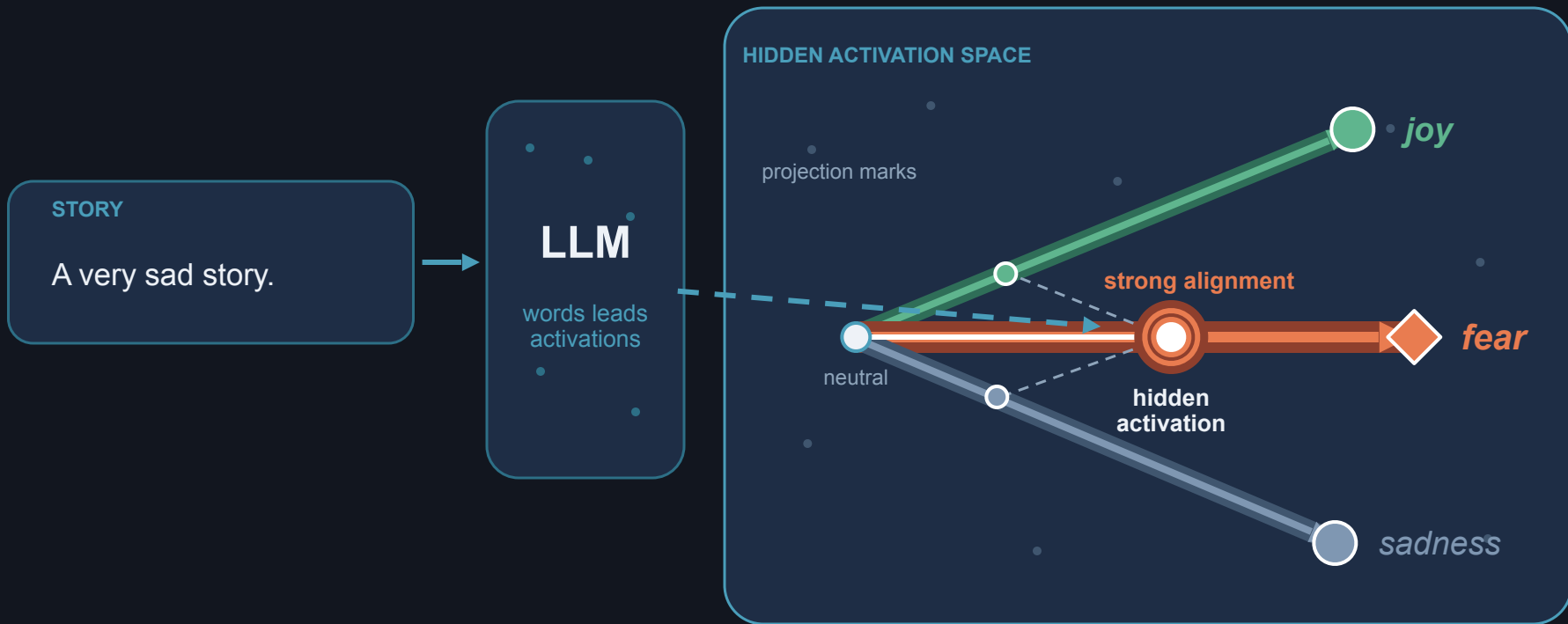
Emotion concepts become directions



Emotion concepts become directions



Emotion concepts become directions



If emotion concepts form directions, could **warmth** and **competence** do the same?

What if social perception is encoded as geometry inside the model?

200 short stories · nameless protagonist · 4 conditions:

High Warmth

Low Warmth

High Competence

Low Competence

Nameless, gender-neutral protagonist in every story, so the vector encodes behaviour, not demographic identity.

Mean difference between high and low conditions gives the direction vector $\mathbf{v}$

If the direction generalizes to new topics never seen during training, the representation is not an artifact of the training stimuli.

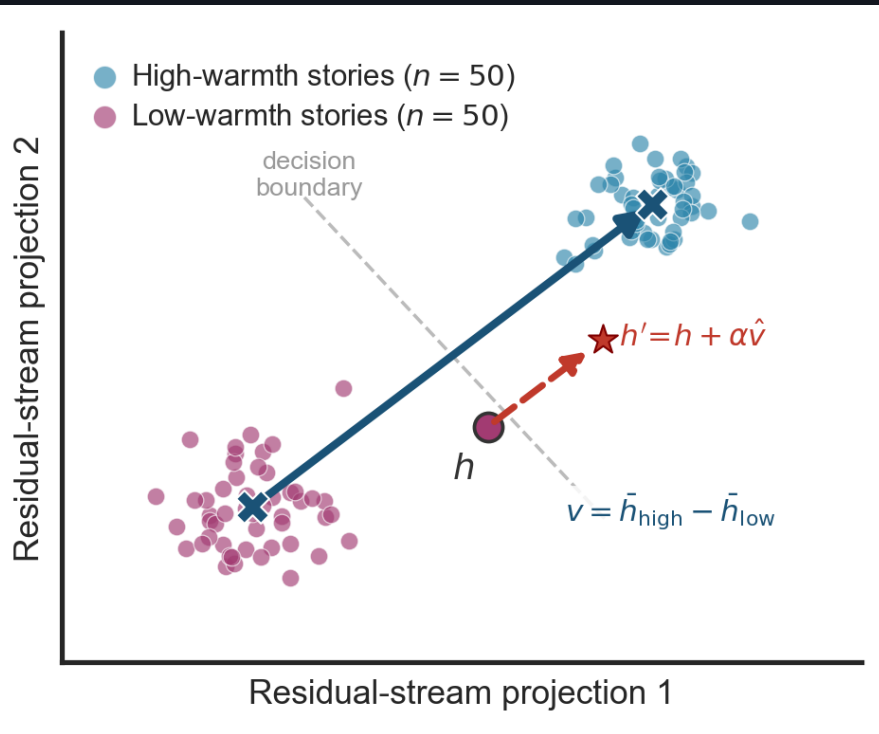


Fig: Concept direction geometry (schematic)

Finding 1: Warmth and competence are linearly encoded in all four models

100%

accuracy, including on
entirely new situations

ρ 0.76–0.99

cross-model story agreement
(4 models, 3 companies)

Gemma-3-12B

Gemma-3-27B

Llama-3.1-8B

Qwen3-14B

Four independently trained models recover the same ordering.

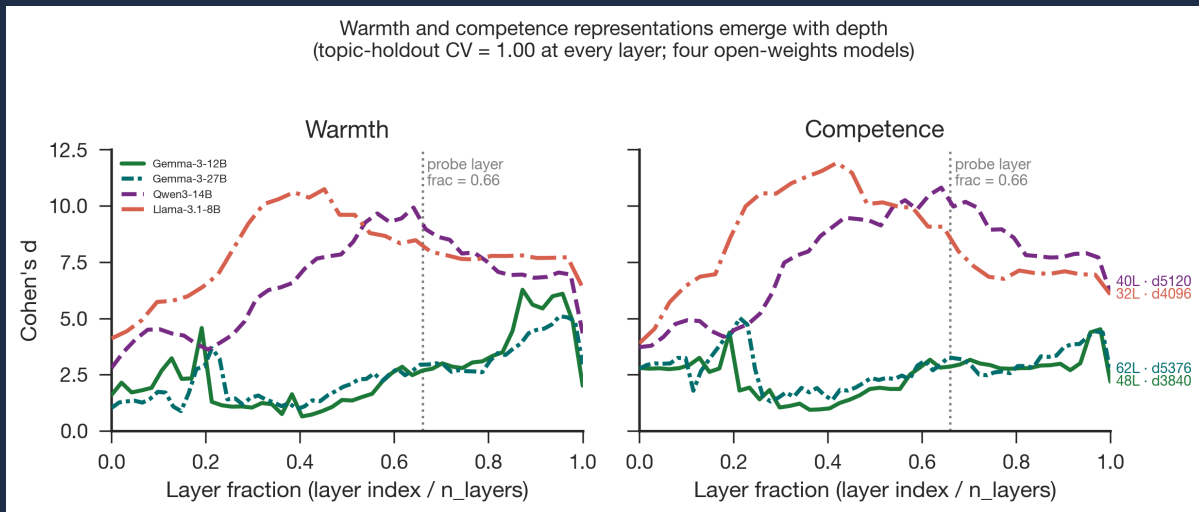


Fig: Layer emergence, Cohen's d by depth

Finding 2: We can steer these directions and flip hiring decisions

Injecting the warmth direction during resume evaluation shifts the callback recommendation, crossing the Yes/No boundary for every name tested.

Caveat: 12B measurement-limited (bf16)

- Baseline: starting margin
- Arrow: steering shift
- Dashed line: decision boundary
- 27B: narrow causal window

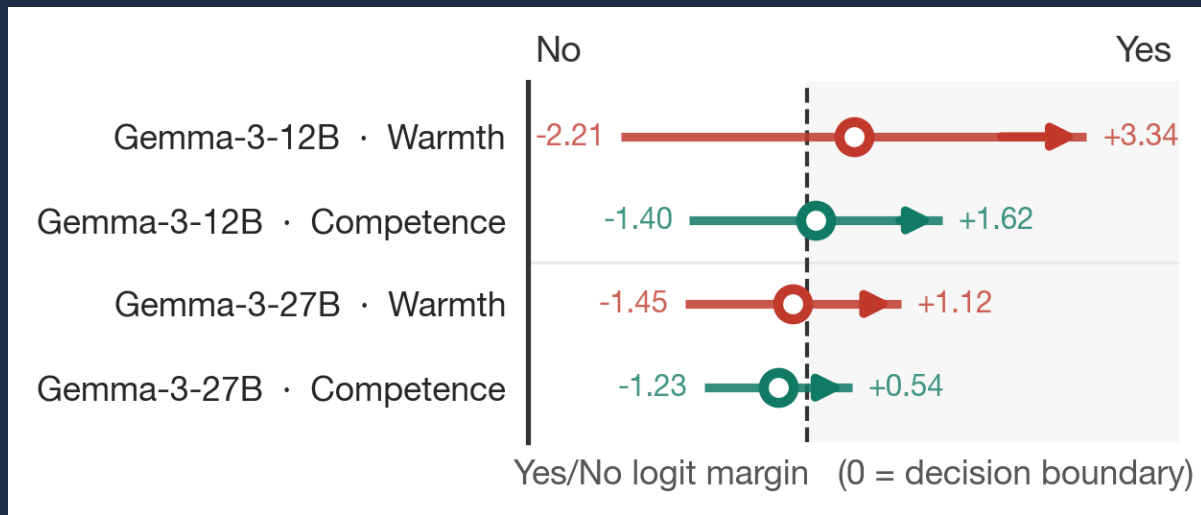


Fig: Diverging steering, Gemma-3-12B and 27B

Finding 3: Model disparities run opposite to the human benchmark

Human benchmark (Gallo & Hausladen, 2024):

White names receive more callbacks than Black names
Males receive slightly more callbacks than females

One possible explanation: safety fine-tuning may have overcorrected for racial associations, reversing the gap rather than removing it.

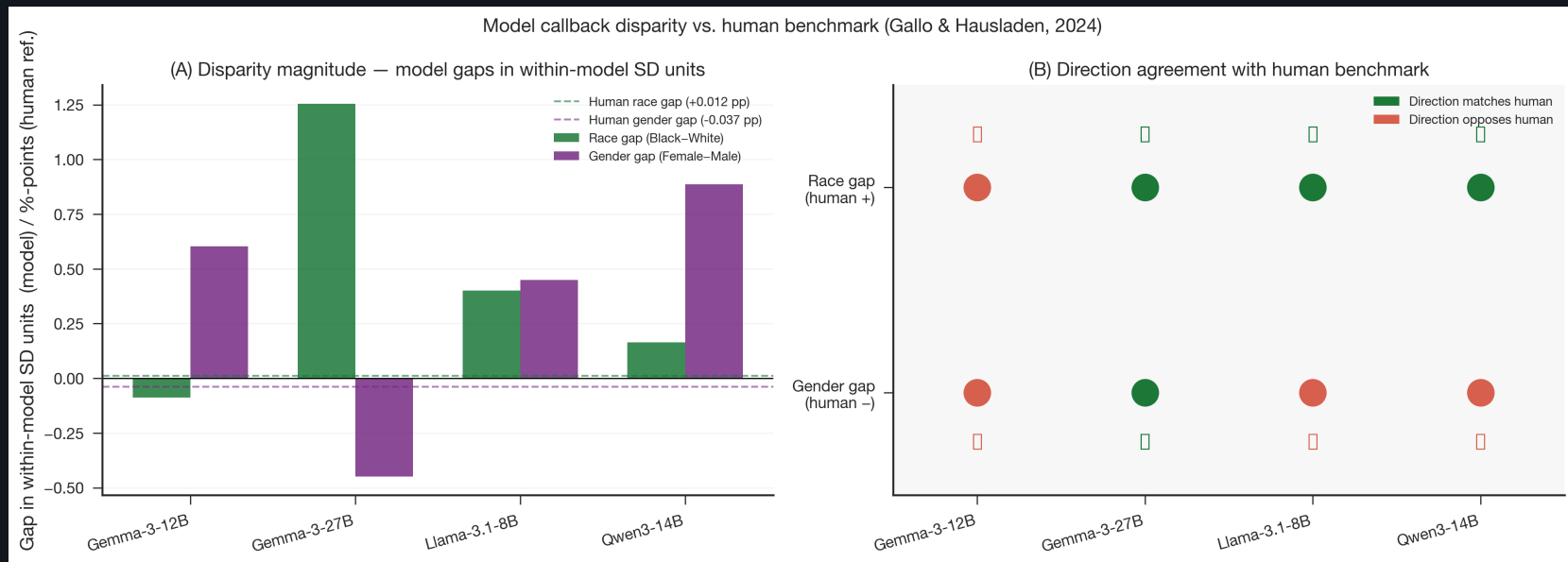


Fig: Model disparity vs human benchmark

The representation mediates the decision, but not where you would expect

Does the warmth/competence probe score carry the name-group signal into the callback decision?

name group → **probe score** → **callback margin**

Significant for Llama (race × warmth). Not significant for Gemma.

Steerability Paradox

Gemma-12B: easiest to steer externally → no significant mediation

Llama-8B: hardest to steer externally → strongest mediation effect

Concept-level steerability and natural causal mediation measure different things.

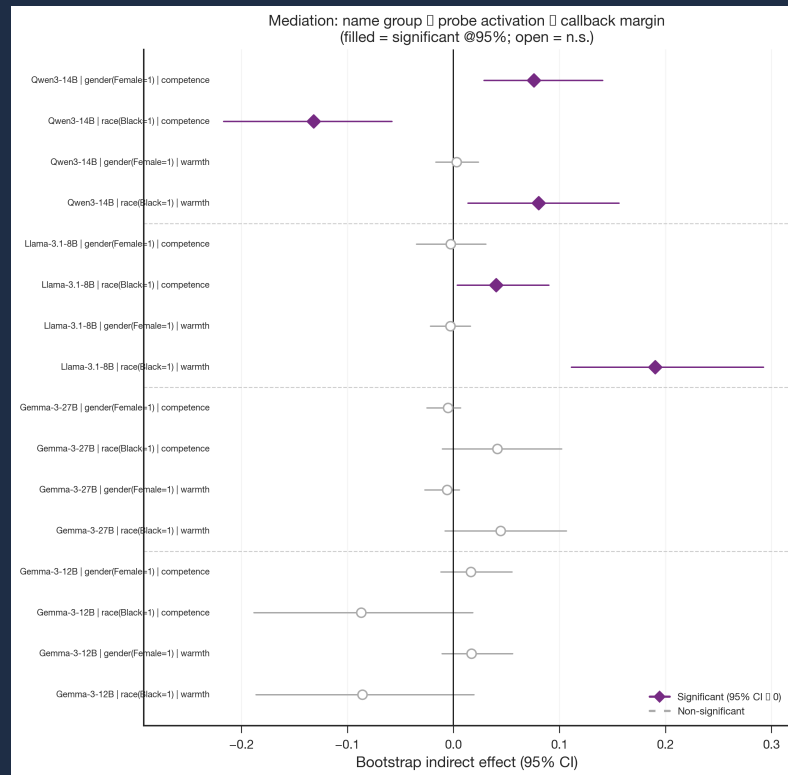


Fig: Bootstrap mediation forest plot, indirect effects with 95% CI

What this means

01

The representations are real

LLMs encode warmth and competence in a structure consistent with human social perception: linearly separable, causally functional, and consistent across four architectures.

02

Disparities persist, but point in the opposite direction

Name-driven hiring disparities remain. Across the models with reliable data, the racial gap reversed direction compared to the human baseline and grew in magnitude.

03

You need both probing and behavioral audits

Internal representation probing and output-level disparity measurement capture different things. Neither alone is sufficient.